

Easy Level (1-20)

1. Write a program to check if a number is even or odd.

```
In [5]: num = int(input("Enter a number:"))
        if num % 2 == 0:
            print ("Even Number")
        else:
            print ("Odd number")
```

Odd number

2. Check if a person is eligible to vote (age 18 or above).

first option

```
In [7]: age = int(input("How old are you?"))
        if age>=18:
            print ("Congratulation, you are eligible to vote.")
        else:
            print ("You are still young to vote!")
```

You are still young to vote!

Second option - using f-string

```
In [11]: age = int(input("How old are you?"))
         if age>=18:
             print ("Congratulation, you are eligible to vote.")
         else:
             i = (18-age)
             print ("You are still young to vote!")
             print(f>Please wait for {i} year(s) to vote.")
```

You are still young to vote!

Please wait for 10 year(s) to vote.

3. Determine if a given year is a leap year or not.

```
In [17]: year = int(input("Enter year to check if it is a leap year:"))
        if year % 4 == 0:
            print ("It is a leap year")
        else:
            print ("It is not a leap year")
```

It is not a leap year

4. Check if a number is positive, negative, or zero.

```
In [24]: num = int(input("Enter a number:"))
        if num == 0:
            print ("You entered 0")
        elif num > 0:
            print ("You entered positive number")
        else:
            print ("You entered negative number")
```

You entered positive number

5. Write a program to find the greatest of two numbers.

```
In [31]: num1 = int(input("Enter first number:"))
        num2 = int(input("Enter second number:"))

        if num1 < num2:
            print(f"{num2} is the greatest number among two.")
        else:
            print(f"{num1} is the greatest number among two.")
```

455 is the greatest number among two.

6 Determine if a number is a multiple of 5.

```
In [34]: num = int(input("Enter a number:"))

        if num % 5 == 0:
            print("The number you entered is a multiple of 5")
        else:
            print("The number you entered is not a multiple of 5")
```

The number you entered is not a multiple of 5

7. Check if a character is a vowel or consonant.

First option

```
In [39]: char = input("Enter a character:")
if char in 'aeiouAEIOU':
    print ("You entered vowel")
else:
    print ("You entered consonant")
```

You entered vowel

Second option

```
In [41]: char = input("Enter a character:").upper()
if char in 'AEIOU':
    print ("You entered vowel")
else:
    print ("You entered consonant")
```

You entered vowel

```
In [44]: char = input("Enter a character:").upper()
if char in 'AEIOU':
    print (f"You entered {char} which is vowel")
else:
    print (f"You entered {char} wchich is consonant")
```

You entered E which is vowel

Third option

```
In [47]: char = input("Enter a character:")
if char in 'AEIOUaeiou':
    print (f"You entered {char} which is vowel")
else:
    print (f"You entered {char} wchich is consonant")
```

You entered t wchich is consonant

8. Determine if a person is eligible for a senior citizen discount (age 60+).

```
In [49]: age = int(input("What is your age?"))
if age >60:
```

```
print ("You are eligible for a senior citizen discount.")
else:
    print("You are NOT eligible for a senior citizen discount.")
```

You are NOT eligible for a senior citizen discount.

9. Write a program to check if a number is a single-digit number.

first option

```
In [57]: import math

num = input("Enter a number:")
if len(num)<=1:
    print("You entered a single digit number")
else:
    print("You entered more than one digit numbers")
```

You entered more than one digit numbers

Second option

```
In [60]: import math

num = input("Enter a number:")
num1 = len(num)
if len(num)<=1:
    print(f"You entered {num1} digit number")
else:
    print(f"You entered {num1} digit numbers")
```

You entered 2 digit numbers

10. Print "Good Morning" if the time is before 12 PM, otherwise print "Good Afternoon".

first option

```
In [71]: time = int(input("What time it is? Enter number using 24 hrs clock:"))
if time < 12:
    print("Hello Sir/Madam, Good morning!")
else:
    print("Hello Sir/Madam, Good afternoon!")
```

Hello Sir/Madam, Good morning!

Second option

```
In [78]: time = int(input("What time it is? Enter number using 24 hrs clock:"))

if time < 12:
    print("Hello Sir/Madam, Good morning!")
elif time < 17:
    print("Hello Sir/Madam, Good afternoon!")
elif time < 21:
    print("Hello Sir/Madam, Good evening!")
else:
    print("Hello Sir/Madam, Good night!")
```

Hello Sir/Madam, Good afternoon!

11. Check if a string is empty or not.

```
In [82]: string = input("Enter a string:")
if string.strip():
    print ("String is NOT empty")
else:
    print ("String is empty")
```

String is empty

12. Verify if a number is a perfect square.

```
In [84]: import math

num = int(input("Enter a number, I can tell you if it is a perfect square:"))
if math.isqrt(num) ** 2 == num:
    print (f"{num} is a perfect square number.")
else:
    print(f"{num} is NOT a perfect square number.")
```

24 is NOT a perfect square number.

13. Determine if a number is between 1 and 100.

```
In [87]: num = int(input("Enter a number:"))
if num <= 100:
    print ("You entered a number that is between 1 and 100")
else:
    print ("You entered a number that is NOT between 1 and 100")
```

You entered a number that is NOT between 1 and 100

14. Print "Weekend" if the day is Saturday or Sunday; otherwise, print "Weekday".

First option

```
In [107... day = str(input("Enter a day:")).capitalize()
if day in ('Saturday, Sunday'):

    print (f"{day} is a weekend, enjoy!")
else:
    print (f"{day} is a weekday, work hard!")
```

Tuesday is a weekday, work hard!

Second option

```
In [115... day = str(input("Enter a day:")).capitalize()
if day in ('Saturday, Sunday'):
    print (f"{day} is a weekend, enjoy!")
elif day in ('Monday, Tuesday, Wednesday, Thursday, Friday'):
    print (f"{day} is NOT a weekend, work hard!")
else:
    print (f"{day} is NOT part of the weekday, check the spelling that you entered!")
```

Monday is NOT a weekend, work hard!

15. Find if a given number is exactly divisible by both 3 and 7.

```
In [121... import math

num = int(input("Enter a number, I can tell you if the number is going to be exactl
if num % 3 == 0:
    print ("It is exactly divisible by 3")
elif num % 7 == 0:
    print ("It is exactly divisible by 7")
else:
    print(f"{num} is NOT exactly divisible either by 3 or 7.")
```

55 is NOT exactly divisible either by 3 or 7.

16. Check if the sum of two numbers is greater than 100.

In [124...

```
import math
num1 = int(input("Enter first number:"))
num2 = int(input("Enter second number:"))

if num1 + num2 > 100:
    print (f"The sum of {num1} and {num2} is greather than 100")
else:
    print(f"The sum of {num1} and {num2} is less than 100")
```

The sum of 74 and 88 is greather than 100

In [127...

```
import math
num1 = int(input("Enter first number:"))
num2 = int(input("Enter second number:"))
num3 = num1 + num2
if num3 > 100:
    print (f"The sum of {num1} and {num2} is {num3} which is greather than 100")
else:
    print (f"The sum of {num1} and {num2} is {num3} which is less than 100")
```

The sum of 10 and 74 is 84 which is less than 100

17. Write a program to find the minimum of two numbers.

first option

In [129...

```
num1 = int(input("Enter first number:"))
num2 = int(input("Enter second number:"))

if num1 < num2:
    print (f"{num1} is the minimum of - {num1} and {num2} numbers")
else:
    print (f"{num2} is the minimum of - {num1} and {num2} numbers")
```

44 is the minimum of - 44 and 55 numbers

Second option

In [132...

```
import math

num1 = int(input("Enter first number:"))
num2 = int(input("Enter second number:"))

num3 = min (num1, num2)
print (f"The minimum number is {num3}")
```

The minimum number is 10

18. Check if a number is divisible by 2 but not by 3.

first option

```
In [138... num = int(input("Enter a number:"))
if num % 2 == 0:
    print (f"{num} is divisible by 2")
else:
    print (f"{num} is not divisible by 2")
```

24 is divisible by 2

Second option

```
In [141... num = int(input("Enter a number:"))

if num % 2 == 0 and num % 3 != 0:
    print(f"{num} is divisible by 2 but not by 3")
elif num % 3 == 0 and num % 2 != 0:
    print(f"{num} is divisible by 3 but not by 2")
elif num % 2 == 0 and num % 3 == 0:
    print(f"{num} is divisible by both 2 and 3")
else:
    print(f"{num} is not divisible by 2 or 3")
```

41 is not divisible by 2 or 3

19. Determine if a given alphabet is uppercase or lowercase.

```
In [146... char = str(input("Enter a alphabet so that I can find out if it is a uppercase or l

if char.isupper():
    print ("You entered uppercase alphabet.")
else:
    print("You entered lowercase alphabet.")
```

You entered lowercase alphabet.

20. Check if a triangle is valid given three side lengths (sum of any two sides must be greater than the third).

In [154...

```
x = int(input("Enter first side: "))
y = int(input("Enter second side: "))
z = int(input("Enter third side: "))

if (x + y > z) and (x + z > y) and (y + z > x):
    print("It is a valid triangle (sides).")
else:
    print("It is not a valid triangle (sides).")
```

It is not a valid triangle (sides).

Check if a triangle is valid given three side lengths (sum of any two angles must be greater than the third).

In [157...

```
x = int(input("Enter first angle: "))
y = int(input("Enter second angle: "))
z = int(input("Enter third angle: "))

if x > 0 and y > 0 and z > 0 and (x + y + z) == 180:
    print("It is a valid triangle (angles).")
else:
    print("It is not a valid triangle (angles).")
```

It is a valid triangle (angles).

This is the end of Easy Level (1-20)

In []: